[DOCUMENT 02 • COMPLETE FEATURE CATALOG & TECHNICAL SPECIFICATIONS]

SignalRescue AI: Feature Name & Architecture Reference

Comprehensive breakdown of all 21 production features across the Android Offline-First Mobile Application (13 screens) and the Government Rescue Command Center Web Platform (10 screens).

11 Mobile	10 Web	E2EE	93 / 93
Field App Features	Command Features	X25519 + AES-GCM	Automated Tests
100% Offline-First SQLite	Real-Time GIS & AI Triage	Zero-trust multi-hop relay	Verified crypto, mesh & UI

01 Mobile Application Features (Citizen & Field Responder App)

ID & Feature Name	Technical Mechanism & Protocol	User Benefit & Operational Value
M-01: Dual-Channel Offline Mesh Engine	Combines **BleTransport** (<4 KB control/SOS frames via Nearby Cluster with lexicographical tie-breaker) and **WifiDirectTransport** (high-throughput P2P streams).	Automatically discovers and connects nearby phones without internet, SIM card, or router; eliminates '0 online' handshake collisions.
M-02: Zero-Trust End-to-End Encryption (E2EE)	Asymmetric **X25519** ECDH key exchange, **HKDF-SHA256** key derivation, **AES-256-GCM** authenticated encryption, and **Ed25519** digital signatures.	Separates cleartext *RoutingHeader* from *EncryptedPayload* (`RNE\x01`). Relays B & C forward packets without reading messages or coordinates.
M-03: Store-and-Forward DTN Outbox & 5-Stage Ticks	Save-Before-Send SQLite guarantee (`encrypted_mesh_packets`), Bloom filter + SQLite deduplication, TTL=5 hop decay, and signed delivery ACKs.	Honest real-time delivery states: *[Saved on this phone]*, *[Queued in outbox]*, *[Relayed by X devices]*, and *[Reached recipient]*.
M-04: Tactile 3-Second Hold SOS Beacon	56dp+ glove-friendly red beacon with 3-second hold-to-confirm guard and structured category picker (Medical, Trapped, Fire, Flood, Missing Person).	Prevents accidental pocket triggers during panic while broadcasting high-priority SOS packets across all mesh peers and gateways.
M-05: Probabilistic WLS Multilateration Engine	10-step Weighted Least-Squares solver using 3-5 peer anchors (BLE RSSI log-distance path loss + Wi-Fi RTT), residual outlier pruning (>2.5x median).	Computes victim position when indoor/rubble blocks GPS, displaying honest uncertainty radius (*Estimated Location • ±35m • 4 GPS anchors*).
M-06: Resumable 16 KB Chunked Image Transfer	Encrypts photos and slices them into 16 KB `RKCH` binary chunks with SHA-256 checksums stored in SQLite `attachment_chunks`.	If a moving peer disconnects at chunk 4/10, transfer automatically resumes from chunk 5 upon reconnection without restarting.
M-07: Offline-First Email Identity & Cloud Sync	Uses sanitized email as unique primary key in local SQLite `user_profile` (`UserAuthService`) with non-blocking background FastAPI sync.	Users can log in offline across app restarts and automatically synchronize their profile and medical notes when internet returns.
M-08: Mesh Group Chat & Invite Protocol	Dedicated `groups`, `group_members`, and `group_invites` SQLite tables with `GROUP_INVITE` and `GROUP_INVITE_RESPONSE` mesh packets.	Families and neighborhood rescue squads can create private groups offline, invite nearby peers, and accept/decline invitations.
M-09: On-Device Offline Survival AI Assistant	Local emergency RAG knowledge base covering water purification, trauma first aid, hypothermia, fracture splinting, and flood evacuation.	Provides instant, step-by-step life-saving medical and survival instructions with zero internet connection required.
M-10: Interactive Offline OpenStreetMap & Shelters	Integrated `flutter_map` with dark/light tile caching, pulsing SOS markers, peer cluster badges, and verified shelter directory with capacities.	Displays nearby mesh peers, active SOS alerts, and safe evacuation shelters even when standalone satellite GPS lock is pending.
M-11: Missing Person Registry & Low-Power OLED Mode	Offline photo/description report builder + 3-tier theme engine (Daylight `#FFFCEB`, Night Navy `#0B1220`, True Black OLED `#000000`).	Helps reunite separated family members over the mesh while conserving up to 40% battery life on AMOLED screens during multi-day blackouts.

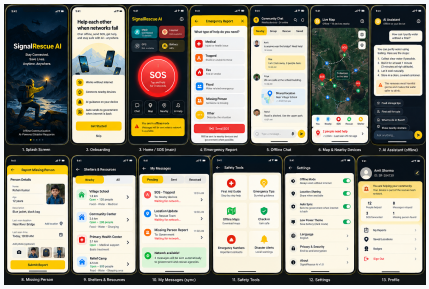


Figure 2.1: Complete 13-Screen SignalRescue AI Mobile Application Suite — Designed with the Hi-Vis Beacon System (Signal Yellow, SOS Red, Beacon Teal).

02 Web Command Center Features (Government & NDRF/SDRF Platform)

Built with React, TypeScript, Tailwind CSS, Leaflet GIS, and Recharts backed by a FastAPI + Dual-Persistence (SQLite + MongoDB Atlas) server, the Command Center turns raw mesh packets into actionable rescue intelligence.

ID & Feature Name	Technical Mechanism & Algorithm	Command & Operational Value
W-01: Real-Time Situational Overview Dashboard	Live WebSocket (`/ws/mesh`) + TanStack Query feeding 4 KPI cards (Active Users, Critical Incidents, Connected Devices, Messages Relayed) & Recharts Donut.	Gives disaster commissioners an instant, zero-latency pulse of the entire regional emergency network.
W-02: Interactive Multi-Layer GIS Live Map	Three-zone tactical Leaflet canvas with custom SVG markers (`SOSMarker`, `FloodZonePolygon`, `RescueTeamMarker`, `CommunicationGapPolygon`).	Allows operators to toggle Users, Mesh Links, Incidents, Rescue Units, and Hazard Zones with one click.
W-03: Spatio-Temporal DBSCAN Incident Clustering	Density-Based Spatial Clustering of Applications with Noise (`/api/v1/incidents/cluster`) grouping nearby SOS packets within epsilon radius.	Condenses 200+ individual panic messages from the same flooded neighborhood into a single verified Incident Polygon with casualty counts.
W-04: Deep AI Explainability & Priority Scoring	Transparent multi-factor severity model outputting confidence % and explicit factor weights (casualty count, time since contact, hazard proximity, battery).	Eliminates 'black-box AI' distrust—commanders see exactly why Incident INC-2025-001 is ranked 94% Critical.
W-05: Communication Black-Hole Anomaly Detector	Monitors P2P heartbeat density across sectors (`/api/v1/anomalies/black-holes`) to flag sudden >75% telemetry drops.	Automatically highlights cut-off 'silence zones' in hatched purple polygons so devastated villages are never overlooked.
W-06: MILP Optimal Fleet Dispatch Solver	Mixed-Integer Linear Programming optimization (`/api/v1/dispatch/optimize`) matching NDRF boats, 4WDs, ambulances, and helicopters to incidents.	Maximizes lives saved per minute by accounting for unit capacity, travel time, capability match, and incident urgency.
W-07: Mesh Device Topology & Gateway Telemetry	Tracks every field node (`SR-XXXX`), battery percentage bar, role (`Online`, `Relay`, `Offline`), and internet gateway status.	Identifies low-battery critical relay nodes before mesh bridges collapse.
W-08: Multi-Channel Field Communications Stream	Filtered real-time feed (All, SOS Alerts, Incident Reports, Medical, Team, System Logs) with hop count and delivery verification badges.	Lets dispatchers read incoming field reports and broadcast targeted updates back into the mesh.
W-09: Dual-Persistence Storage (SQLite + MongoDB)	Transactional local SQLite (`gateway_telemetry.db`) paired with asynchronous TLS-secured MongoDB Atlas replication.	Ensures the local command post continues operating if government WAN links flicker, while syncing globally to cloud headquarters.
W-10: OASIS CAP v1.2 XML & CAD Interoperability	One-click export of Common Alerting Protocol v1.2 XML (`/api/v1/interop/cap/feed.xml`), NG911 CAD JSON, and CSV reports.	Integrates directly with national Disaster Management Authorities (NDMA), WMO, and legacy dispatch consoles.

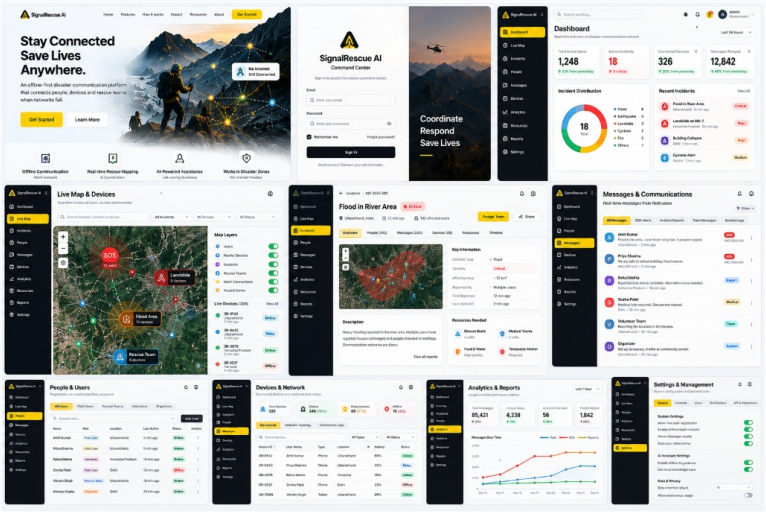


Figure 2.2: Complete 10-Screen SignalRescue AI Web Command Center — From Public Landing & Secure Login to Live GIS Map, Incident AI Explainability, Devices, and Analytics.